$\text{lam}\,(v1\,\$\,v0) = v0$

$\Rightarrow \eta : \underbrace{t}_{:\,Tm\,\Gamma\,(A\Rightarrow B)} = \text{lam}\,(\underbrace{t[p]\,\$\,v0}_{:\,Tm\,(\Gamma\triangleright A)\,B})$

Véges típusok, $0, 1, +, \otimes$

Szorzat, tuple, record, class

$_\times_ : Ty \to Ty \to Ty$

$(\text{fst}\ t)[\sigma] = \text{fst}\,(t[\sigma])$

$(u,v)[\sigma] = (u[\sigma], v[\sigma])$

$\text{fst}_{A,B}, \text{snd} : Tm\,\Gamma\,(A\times B) \cong Tm\,\Gamma\,A \times Tm\,\Gamma\,B : _,_$

$\times\beta_1 : \text{fst}(u,v) = u$

$\times\beta_2 : \text{snd}(u,v) = v$

$\Rightarrow\beta : \text{lam}\ t\,\$\,u = t[\langle u\rangle]$

$\times\eta : t = (\text{fst}\ t, \text{snd}\ t)$

$v0 : Tm\,(\diamond\triangleright(Nat\times Bool)\times String)\ ((Nat\times Bool)\times String)$

$\text{fst}_{Nat\times Bool,\,String}(v0) : Tm\,(\diamond\triangleright(Nat\times Bool)\times String)\ (Nat\times Bool)$

$\text{fst}_{Nat,\,Bool}(\text{fst}\ v0) : Tm\,(\diamond\triangleright(Nat\times Bool)\times String)\ Nat$

$\text{lam}(\text{fst}(\text{fst}\ v0)) : Tm\,\diamond\,((Nat\times Bool)\times String \Rightarrow Nat)$

$\text{lam}(\text{fst}(\text{fst}\ v0))\,\$\,((m1, hu), \text{"hello"}) = \text{fst}(\text{fst}\ v0)[\langle (m1, hu), \text{"hello"}\rangle] =$

$\text{fst}(\text{fst}\ v0[\langle_\rangle]) = \text{fst}(\text{fst}(v0[\langle_\rangle])) = \text{fst}(\text{fst}((m1, hu), \text{"hello"})) \overset{\times\beta_1}{=}$

$\text{fst}(m1, hu) \overset{\times\beta_1}{=} m1$

$A, B : Ty$

$A\times B \cong B\times A$

$(A\times B)\times C \cong A\times(B\times C)$

$Bool\times Nat \cong Nat\times Bool$

$t := \text{lam}(\text{snd}_{Bool,Nat}\ v0, \text{fst}_{Bool,Nat}\ v0)$

$t' := \text{lam}(\text{snd}_{Nat,Bool}\ v0, \text{fst}_{Nat,Bool}\ v0)$

$A \cong B := (t : Tm\,\diamond\,(A\Rightarrow B)) \times (t' : Tm\,\diamond\,(B\Rightarrow A)) \times (t[p]\,\$\,(t'[p]\,\$\,v0) = \underbrace{v0}_{:\,Tm\,(\diamond\triangleright A)\,B}) \times (t'[p]\,\$\,(t[p]\,\$\,v0) = \underbrace{v0}_{:\,Tm\,(\diamond\triangleright B)\,B})$

$:\,Tm\,(\diamond\triangleright A)\,A$

$Tm\,(\diamond\triangleright A)\,B := (Tm\,A \to Tm\,B)$

$Tm\,(\diamond\triangleright A\triangleright B\triangleright C)\ D := (Tm\,A\to Tm\,B\to Tm\,C\to Tm\,D)$

$t[p]\,\$\,(t'[p]\,\$\,v0) =$

$t[p]\,\$\,(\text{lam}(\text{snd}\,v0, \text{fst}\,v0)[p]\,\$\,v0) \overset{\text{lam}[\,]}{=}$

$t[p]\,\$\,(\text{lam}((\text{snd}\,v0, \text{fst}\,v0)[p^+])\,\$\,v0) \overset{,[\,],\ \text{fst}[\,],\ \text{snd}[\,]}{=}$

$t[p]\,\$\,(\text{lam}(\text{snd}(v0[p^+]), \text{fst}\,v0[p^+])\,\$\,v0) \overset{v0[+]}{=}$

$t[p]\,\$\,(\text{lam}(\text{snd}\,v0, \text{fst}\,v0)\,\$\,v0) \overset{\Rightarrow\beta}{=}$

$t[p]\$((\mathrm{snd}\,v0, \mathrm{fst}\,v0)[\langle v0\rangle]) \overset{\cdot[]}{=}$

$t[p]\$(\mathrm{snd}\,v0[\langle v0\rangle], \mathrm{fst}\,v0[\langle v0\rangle]) \overset{\mathrm{fst}[], \mathrm{snd}[]}{=}$

$t[p]\$(\mathrm{snd}(v0[\langle v0\rangle]), \mathrm{fst}(v0[\langle v0\rangle])) \overset{v0[\langle\rangle]}{=}$

$t[p]\$(\mathrm{snd}\,v0, \mathrm{fst}\,v0) =$

$\mathrm{lam}(\mathrm{snd}\,v0, \mathrm{fst}\,v0)[p]\$(\mathrm{snd}\,v0, \mathrm{fst}\,v0) \overset{\cdots}{=}$

$\mathrm{lam}(\mathrm{snd}\,v0, \mathrm{fst}\,v0)\$(\mathrm{snd}\,v0, \mathrm{fst}\,v0) \overset{\Rightarrow\beta}{=}$

$(\mathrm{snd}\,v0, \mathrm{fst}\,v0)[\langle \mathrm{snd}\,v0, \mathrm{fst}\,v0\rangle] \overset{\cdot[], \mathrm{snd}[], \mathrm{fst}[], v0[\langle\rangle]}{=}$

$(\mathrm{snd}(\mathrm{snd}\,v0, \underline{\mathrm{fst}\,v0}), \mathrm{fst}(\mathrm{snd}\,v0, \mathrm{fst}\,v0)) \overset{\times\beta_1, \times\beta_2}{=}$

$(\mathrm{fst}\,v0, \mathrm{snd}\,v0) \overset{\times\eta}{=}$

V0

$\mathrm{Ty}, \times$ Abel félcsoportot alkotnak

egységelem, unit

$\mathrm{Unit} : \mathrm{Ty}$

$\mathrm{triv} : \mathrm{Tm}\,\Gamma\,\mathrm{Unit}$

$\mathrm{Unit}\eta : (t : \mathrm{Tm}\,\Gamma\,\mathrm{Unit}) \to t = \mathrm{triv}$

$\mathrm{Tm}\,\Gamma\,(A\times B) \cong \mathrm{Tm}\,\Gamma\,A \times \mathrm{Tm}\,\Gamma\,B$

$\mathrm{Tm}\,\Gamma\,\mathrm{Unit} \cong T$

$\cancel{\mathrm{Tm}\,\Gamma\,(A+B) \cong \mathrm{Tm}\,\Gamma\,A + \mathrm{Tm}\,\Gamma\,B}$

$\mathrm{Tm}\,(\Gamma \triangleright A+B)\,C \cong \mathrm{Tm}\,(\Gamma\triangleright A)\,C \times \mathrm{Tm}\,(\Gamma\triangleright B)\,C$

case switch

$\mathrm{Tm}\,\Gamma\,(A\Rightarrow B) \cong \mathrm{Tm}\,(\Gamma\triangleright A)\,B$

$\mathrm{Ty}, \mathrm{Unit}, \times$ kommutatív monoid

$\mathrm{Unit}\times A \cong A\times\mathrm{Unit} \cong A$

$(A\times B)\times C \cong A\times(B\times C)$

$A\times B \cong B\times A$

$_+_ : \mathrm{Ty} \to \mathrm{Ty} \to \mathrm{Ty}$

ömeg, sum, coproduct, enum, leírja az öröklődés egy aspektusát

$A+B$-ben vagy egy A típusú term, vagy egy B típusú term van

data Tm : Set where

$\mathrm{lam} : (\mathrm{Tm}\to\mathrm{Tm}) \to \mathrm{Tm}$

$\mathrm{app} : \mathrm{Tm} \to (\mathrm{Tm}\to\mathrm{Tm})$

not strictly positive

nem algebrai:

$\mathrm{Tm}\times\mathrm{Tm}\to\mathrm{Tm}$

Ötnegy) $\quad Bool := Unit + Unit \qquad 2 = 1+1$

left, right : $Tm(\Gamma \triangleright A+B)\,C \cong Tm(\Gamma \triangleright A)\,C \times Tm(\Gamma \triangleright B)\,C$: case

$+\beta$: case (left t, right t) = t

inl. : $Tm\,\Gamma\,A \to Tm\,\Gamma\,(A+B)$

inl a := $\underbrace{\text{left } v0}_{:\,Tm(\Gamma \triangleright A)(A+B)}\,[\langle a\rangle]$

$+\eta_1$: left (case u v) = u

$+\eta_2$: right (case u v) = v

Nat + Nat $\cong$ Nat × Bool

verőazon. $\quad$ rendelési szám

kikuldetke : $Tm(\Gamma \triangleright Nat+Nat)\,Bool$ $\qquad A+A = 2\times A$

case u v ahol $u : Tm(\Gamma \triangleright Nat)\,Bool$

$v : Tm(\Gamma \triangleright Nat)\,Bool$

case u v [<inl a>] = u [<a>]

case u v [<inr b>] = v [<b>]

kikuldetke := case v0 of

left vers_azon → ....

right rend_szám → ....

left, right : $Tm(\Gamma \triangleright A+B)\,C \cong Tm(\Gamma \triangleright A)\,C \times Tm(\Gamma \triangleright B)\,C$: case

case u v [<left v0 [<a>]>] $\overset{?}{=}$ u[<a>]

||

left (case u v) = u

right (case u v) = v

case (left t) (right t) = t

left t [σ^+] = left (t [σ^+])

right t [σ^+] = right (t [σ^+])

(case u v) [σ^+] = case (u [σ^+]) (v [σ^+])

t : $Tm(\Gamma \triangleright C)\,D$

t [p^+] : $Tm(\Gamma \triangleright A+B \triangleright C)\,D$

t [p^+] [case u v] : $Tm(\Gamma \triangleright A+B)\,D$

t [p^+] [<case u v>] = D

case

$A \xrightarrow{u} C$

$B \xrightarrow{v} C$

$A+B \xrightarrow{\text{case } u\,v} C$

$A \xrightarrow{u} (C \xrightarrow{t} D)$

$B \xrightarrow{v} (C \xrightarrow{t} D)$

$A+B \xrightarrow{\text{case}} D$

w [p^+] [<case u v>] =

case (w [p^+] [<u>])

(w [p^+] [<v>])

w [p^+] [<left t>] =

left (w [p^+] [<t>])

vO [<t>] = t

u x [<t>] = x

x [p] = u x

vO [σ^+] = vO

vS x [σ^+] = x [σ] [p]

t [p] [<u>] = t

t [<u>] [σ] = t [σ^+] [(u [σ])]

t [p] [σ^+] = t [σ] [p]

$D \leftarrow C \leftarrow A+B$

t [p^+] [<case u v>] = case (t [p^+] [<u>]) (t [p^+] [<v>])

A,B,C $\qquad$ A,B,D

$D \leftarrow C \leftarrow A$ $\qquad$ $D \leftarrow C \leftarrow B$